

Spina bifida support

Regenerative biologic adjunct · intramuscular administration

An adjunctive framework intended to support neural signaling, reduce neuroinflammation, and support systemic repair mechanisms alongside standard medical and surgical management. This therapy is not intended to cure spina bifida and is investigational and off-label.

Inclusion

- Diagnosed spina bifida (myelomeningocele, meningocele, or occulta)
- Pediatric or adult patients per clinic policy
- Medically stable without active infection
- Informed consent obtained

Exclusion

- Active systemic infection
- Active malignancy
- Pregnancy unless approved by the Medical Director

Dosing & administration

REGIMEN A · TOTAL

5 cc

EXM 2 cc + PV-MAX 3 cc,
IM

NEEDLE / OBSERVATION

22-25 G

observe 15-30 min post-procedure

REGIMEN B · NOVA PRODUCT DISTRIBUTION (7 cc TOTAL)

- NOVA-M1: 2 cc per glute, bilaterally
- NOVA-E1: 1.5 cc per glute, bilaterally
- Preferred site: gluteal musculature (volume tolerance, systemic absorption)
- Alternative: deltoid, maximum 2 cc per deltoid
- Aspirate prior to injection and inject slowly
- Regimens A and B are alternatives, not additive — confirm which applies before dosing.

OUTCOME MEASURES

NEUROLOGIC STABILITY

PAIN / SPASTICITY

FUNCTIONAL MOBILITY

QUALITY OF LIFE

PREPARATION & HANDLING

- Thaw according to manufacturer specifications
- Glycerol-based products do not require washing
- Use within 15-30 minutes post-thaw
- Maintain strict sterile technique throughout

TREATMENT SCHEDULE

Single initial administration. Repeat dosing may be considered every **4-8 weeks** based on clinical response; re-evaluate after **2-3 sessions**.

ADJUNCTIVE CARE

Continue standard medical and surgical management. Consider nutritional optimization (B-complex, magnesium, omega-3) and physical or neuro-rehabilitation as indicated.

RISKS & SAFETY

Injection-site discomfort, bruising, or transient inflammatory response; potential site infection. No known tumorigenic risk has been identified based on the acellular nature of the biologics.

OVERSIGHT

All treatments must be ordered and supervised by a licensed physician experienced in regenerative medicine, and must comply with applicable state and federal regulations, institutional policy, and informed-consent requirements. No claims of disease cure are made.

PATIENT NAME	DATE OF BIRTH	PATIENT ID / MRN	DATE OF VISIT
EVALUATING PHYSICIAN	INITIAL · FOLLOW-UP VISIT TYPE — CIRCLE	REFERRING PROVIDER, IF ANY	TIME IN / OUT

S SUBJECTIVE

Spina bifida — supportive care visit —
 CHIEF COMPLAINT _____ ONSET / DURATION _____ SEVERITY — VAS 0-10 _____

QUALITY / RADIATION _____ AGGRAVATING FACTORS _____ RELIEVING FACTORS _____

Ongoing: PT · bracing · urology routine · neurosurgery follow-up — confirm
 PRIOR TREATMENT — CIRCLE ALL TRIED · ADD DATES _____

mobility · transfers · bladder/bowel routine · skin checks · daily ... _____ fever · chills · weight loss · neuro changes — note positives
 FUNCTIONAL IMPACT _____ PERTINENT ROS _____

MEDICATIONS — INCL. ANTICOAGULANTS, IMMUNOSUPPRESSANTS _____ ALLERGIES · RELEVANT PMH — CANCER, INFECTION, PREGNANCY _____

O OBJECTIVE

BP	HR	TEMP	WEIGHT	GENERAL APPEARANCE
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Motor level and strength — key muscle groups _____

Tone and reflexes _____

Skin — planned injection sites and pressure areas _____

Mobility — gait / device / wheelchair skills _____

Weight for dosing _____

IMAGING / STUDIES REVIEWED — MODALITY, DATE, FINDINGS	BASELINE — Motor exam	BASELINE — B/B diary	BASELINE — PT goals
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A ASSESSMENT

Spina bifida — physician-directed supportive protocol
 WORKING DIAGNOSIS — CONFIRM OR AMEND _____ ADDITIONAL DIAGNOSES / CODES _____

☐ Appropriate candidate — indications met, contraindications absent	☐ Alternatives discussed, including conservative care and doing nothing
☐ Non-FDA-approved status explained; no outcome promised	☐ Patient questions answered; understanding confirmed

P PLAN

Proposed treatment. IM support, ONE regimen only: Regimen A (EXM 2 cc + PV-MAX 3 cc, gluteal preferred) OR Regimen B (NOVA-M1 + NOVA-E1, 7 cc bilateral gluteal), 22-25 G, weight-based in pediatrics. Observation 15-30 min.

☐ Informed consent (CN-PC) reviewed and signed	☐ Procedure performed today — see CN-OP-SB · or scheduled (date right)
☐ Fees, non-coverage by insurance, and refund policy reviewed	☐ Follow-up: repeat consideration 4-8 wk; PT continues

PHYSICIAN SIGNATURE	PRINTED NAME & LICENSE NUMBER	PROCEDURE DATE, IF SCHEDULED
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PATIENT NAME

DATE OF BIRTH

PATIENT ID / MRN

DATE OF PROCEDURE

PERFORMING PHYSICIAN

Spina bifida — physician-directed supportive protocol

PRE-PROCEDURE DIAGNOSIS — CONFIRM OR AMEND

START TIME

ASSISTING STAFF

PRE-PROCEDURE CHECKLIST

Check each box before the first draw

☐ Informed consent (CN-PC) signed and on file; SOAP note (CN-SN-SB) complete

☐ Screening reviewed — no contraindications present

☐ Time-out performed — correct patient, site, product, route

☐ Five checks read aloud with second staff: ID · product & lot · expiry · volume & route · consent

☐ Baseline outcome measures recorded: Motor function · bladder/bowel diary · PT goals

☐ Site marked and prepped — chlorhexidine / povidone-iodine (circle); chain-of-custody log initialed

BP · HR · TEMP

agent · concentration · volume — write NONE i...

BP · HR · TEMP

PRE-PROCEDURE VITALS

LOCAL ANESTHETIC

POST-OBSERVATION VITALS

PRODUCTS ADMINISTERED

Regimens A and B are alternatives — select ONE

PRODUCT & TARGET	PLANNED	LOT	EXPIRY	GIVEN
REGIMEN A — EXM + PV-MAX IM, gluteal preferred · EXM 2 cc + PV-MAX 3 cc	5 cc total			
REGIMEN B — NOVA-M1 + NOVA-E1 IM, bilateral gluteal	7 cc total			

TECHNIQUE

Position: prone or side-lying; vastus lateralis in small children

- Circle the regimen used — A and B are never combined.
- Respect IM volume limits: gluteal up to 3 cc per side in adults; split doses across sites, weight-based in pediatrics.
- Aspirate before injecting at every site; inject slowly.

GLUTEAL L · R · BILATERAL — or VASTUS LATERALIS L · R — circle

SITE / SIDE

REGIMEN A · REGIMEN B — circle one

GUIDANCE

22-25 G, per site and weight

NEEDLE USED

target 15-30 min

OBSERVATION — MINUTES

NARRATIVE — FINDINGS, ASPIRATE, DEVIATIONS FROM PROTOCOL, PATIENT RESPONSE

POST-PROCEDURE & DISPOSITION

Tolerated: WELL · WITH EVENTS — circle

Discharged: STABLE · AMBULATORY · WITH ESCORT — circle

☐ Patient note CN-PN-SB reviewed and given at discharge

COMPLICATIONS / ADVERSE EVENTS — WRITE "NONE" IF NONE

Any adverse event also requires the adverse-event response steps in this book.

Follow-up plan. Repeat consideration at 4-8 weeks. Physical therapy, bracing, and specialist care continue unchanged.

FOLLOW-UP SCHEDULED FOR

PHYSICIAN SIGNATURE

PRINTED NAME & LICENSE NUMBER

DATE & END TIME

Template — photocopy per patient and file with the consent form and SOAP note. Investigational and off-label use; not FDA-approved to diagnose, treat, cure, or prevent disease.

11 · PROCEDURE TEMPLATE

Today you received intramuscular injections of cell and exosome products into the hip and buttock muscles. This note is written for the patient and family. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

WHAT TO EXPECT

Changes are gradual — do not judge the result in the first days

FIRST 48 HOURS

Soreness at the injection sites is common, and a low-grade temperature for under 24 hours can occur. Normal activity as tolerated is fine.

FIRST WEEKS

Physical therapy, bracing, and bladder and bowel routines continue exactly as before — this treatment adds to that care, never replaces it.

BY FOLLOW-UP

Bring your notes and diary. Progress is measured against PT goals and the baseline exam, not day-to-day impressions.

DO

- Keep every physical therapy appointment
- Bathe normally after 24 hours
- Use acetaminophen (Tylenol), dosed by weight, for soreness
- Keep the bladder and bowel routine unchanged

AVOID

- Anti-inflammatory pain relievers for 72 hours unless your physician says otherwise
- Hot soaking baths for 24 hours
- Massage of the injection sites for 48 hours
- Judging results in the first days

CALL US RIGHT AWAY IF

For an emergency, call 911 first

- Fever above 101 °F
- Unusual sleepiness or lethargy
- Spreading redness, warmth, or swelling at an injection site
- New weakness, or any change in bladder or bowel function

CLINIC PHONE

AFTER-HOURS PHONE

NOTES FROM YOUR VISIT